

ONLINE SHOP MANAGEMENT SYSTEM



Project Report

CSE282.5 — Introduction to Programming Language II (Java)

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Complex Engineering Project

Submitted

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Team Members

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1. Introduction

The Online Shop Management System is a Java-based desktop application developed as a Complex Engineering Project for the CSE282 — Introduction to Programming Language II course. The project simulates the basic operations of an online shopping system, including product, customer, order, and payment management through a Java Swing graphical user interface.

The system applies the fundamental principles of Object-Oriented Programming (OOP), including Encapsulation, Abstraction, Inheritance, and Polymorphism. It also demonstrates exception handling, stock management, and file-based data persistence.

2. Objectives

2.1 Main Objective

The main objective is to develop a functional Java-based Online Shop Management System that demonstrates the practical application of Object-Oriented Programming concepts.

2.2 Specific Objectives

- Develop a user-friendly graphical interface using Java Swing.
- Manage product information and product stock.
- Register and manage customers.
- Create and manage customer orders.
- Implement different payment methods.
- Demonstrate Encapsulation, Abstraction, Inheritance, and Polymorphism.
- Implement custom exception handling for invalid operations.
- Maintain application data using local files.
- Develop a modular and maintainable Java application.

2.3 Key Functionalities

- Product management
- Customer management
- Order management
- Stock tracking
- Payment processing
- Input validation
- Exception handling
- File-based data storage
- Graphical user interface

3. Motivation

Online shopping has become an important part of modern commerce. Managing products, customers, orders, stock, and payments manually can become difficult as the number of transactions increases.

This project was selected to understand how a real-world shopping system can be represented using programming concepts learned in Java. It also provides practical experience with classes, inheritance, interfaces, polymorphism, exception handling, and file handling in a complete application.

4. Applications

The Online Shop Management System can be used as a simplified model of a small-scale online shopping or retail management application.

4.1 Potential Users

- Shop owners
- Small business operators
- Store managers
- Sales representatives
- Customers
- Students learning Java and OOP concepts

4.2 Practical Uses and Benefits

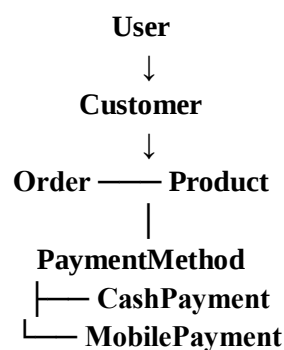
- Maintain product information and track stock.
- Register and manage customer information.
- Manage customer orders.
- Process different types of payments.
- Demonstrate a basic e-commerce workflow.

5. Project Description

5.1 Overall System Description

The system consists of interconnected classes representing the main entities of an online shop. The major entities include User, Customer, Product, Order, PaymentMethod, CashPayment, and MobilePayment. The MainGUI class provides the main graphical interface through which users interact with the system.

Major class relationship:



5.2 Main Features

- Product Management — manages product information, pricing, and available stock.

- Customer Management — registers customers and associates them with orders.
- Order Management — connects customers, products, and payment methods.
- Payment Processing — supports Cash Payment and Mobile Payment.
- Stock Management — validates product availability before transactions.
- Exception Handling — uses custom exceptions such as OutOfStockException.

5.3 Major Classes

Class	Main Responsibility
Product	Stores product information and manages stock
User	Provides common user properties and behavior
Customer	Represents a customer and extends User
Order	Manages customer orders
PaymentMethod	Defines the common payment interface
CashPayment	Implements cash payment
MobilePayment	Implements mobile payment
OutOfStockException	Handles insufficient product stock
MainGUI	Provides the main graphical user interface

5.4 System Workflow

Start → Launch MainGUI → Manage Products → Manage Customers → Create Order → Check Stock → Select Payment Method → Process Order → Update/Store Data → End

5.5 Java and OOP Concepts Used

Encapsulation: Private fields and controlled access through methods such as getters and setters.

Abstraction: The User class and PaymentMethod interface provide common definitions for related operations.

Inheritance: Customer extends User, allowing common user properties and behavior to be reused.

Polymorphism: CashPayment and MobilePayment implement PaymentMethod and can be handled through the common interface.

Exception Handling: Java try-catch mechanisms and the custom OutOfStockException handle invalid operations.

6. Tools & Technologies

Tool / Technology	Purpose
Java	Main programming language
Java Swing	Graphical User Interface
IntelliJ IDEA	Development environment
Git	Version control
GitHub	Source code repository
Local Files	Data persistence

The project uses standard Java functionality and does not require an external database server.

7. Output / Results

The completed system provides a graphical interface through which users can interact with product, customer, order, payment, stock, and exception-handling functionalities.

7.1 Screenshots

Insert screenshots of the following developed system screens in this section:

- Main GUI
- Product Management
- Customer Management
- Order Processing
- Payment Processing
- Error Handling / Out-of-Stock Message

7.2 Sample Error Handling

When a customer attempts to purchase a product that does not have sufficient stock, the system uses the custom `OutOfStockException` to handle the situation instead of allowing an invalid transaction.

8. Contribution

Team Member	Main Contribution	Module / Feature
Muhammad Sydul Islam	Developed Order functionality and worked on overall project integration.	Order
Ainun Nahar Kona	Developed the payment interface and payment implementations.	PaymentMethod, CashPayment, MobilePayment
Saidur Rahman Joy	Developed product information, pricing, and stock management.	Product
Sadia Islam Sinthya	Developed user and customer-related functionality.	User, Customer

All team members contributed to the development and completion of the overall project.

9. Conclusion

The Online Shop Management System was successfully developed as a Java-based desktop application for the CSE282 course. The project achieved its main objectives by implementing product, customer, order, and payment management functionalities within a single system.

The project provided practical experience in applying the four major Object-Oriented Programming principles: Encapsulation, Abstraction, Inheritance, and Polymorphism. It also demonstrated Java Swing GUI development, custom exception handling, and file-based data persistence.

9.1 Limitations

- Uses local file-based storage instead of a database.
- Designed as a desktop application rather than a web or mobile application.
- Limited authentication and security features.
- No real online payment gateway integration.

- Primarily designed for academic demonstration.

9.2 Future Improvements

- Integration with a database such as MySQL.
- Development of a web-based version.
- Secure user authentication.
- Integration with real payment gateways.
- Product search and filtering.
- Improved reporting and sales analytics.
- Administrator dashboard.
- Cloud-based data storage and synchronization.